



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

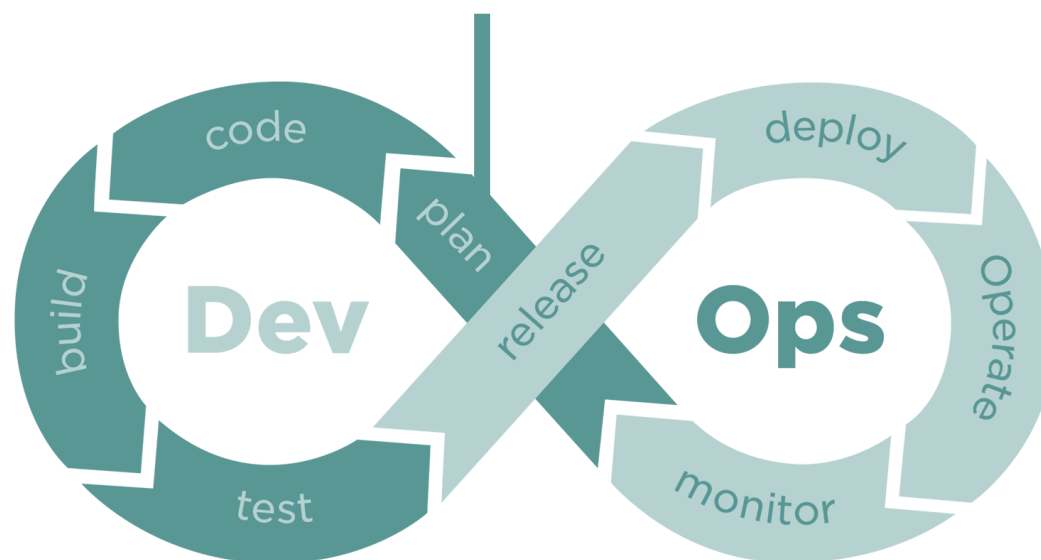
CS304 SOFTWARE ENGINEERING

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WHERE ARE WE NOW?

Requirement analysis





LECTURE 4

- Overview of software requirements
- Stakeholders
- Types of requirements
- Requirements analysis in Scrum



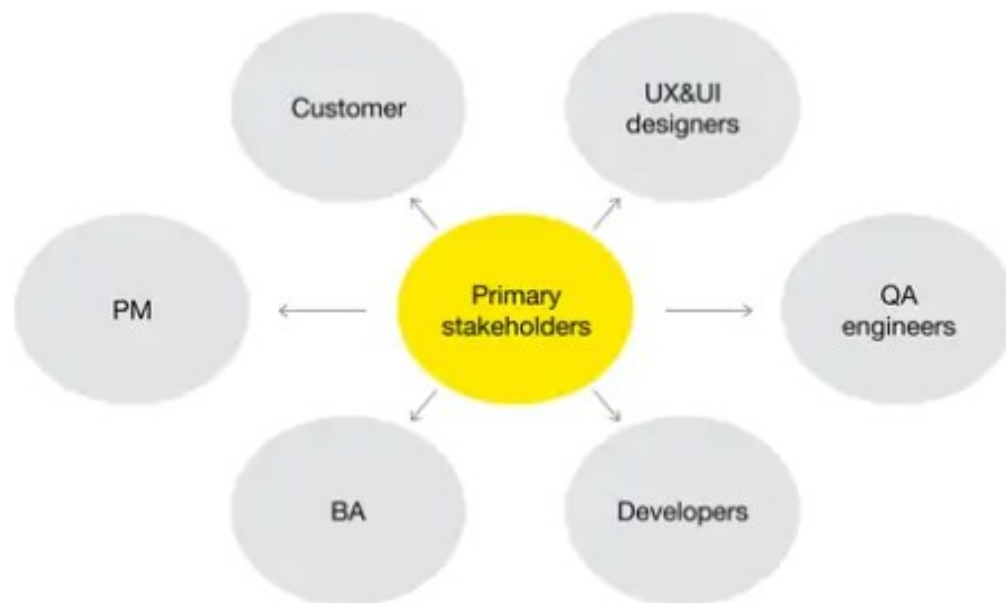
WHAT IS SOFTWARE REQUIREMENT?

- Software requirements (需求) describe the following:
 - what the system should do
 - the services that it provides
 - the constraints on its operation

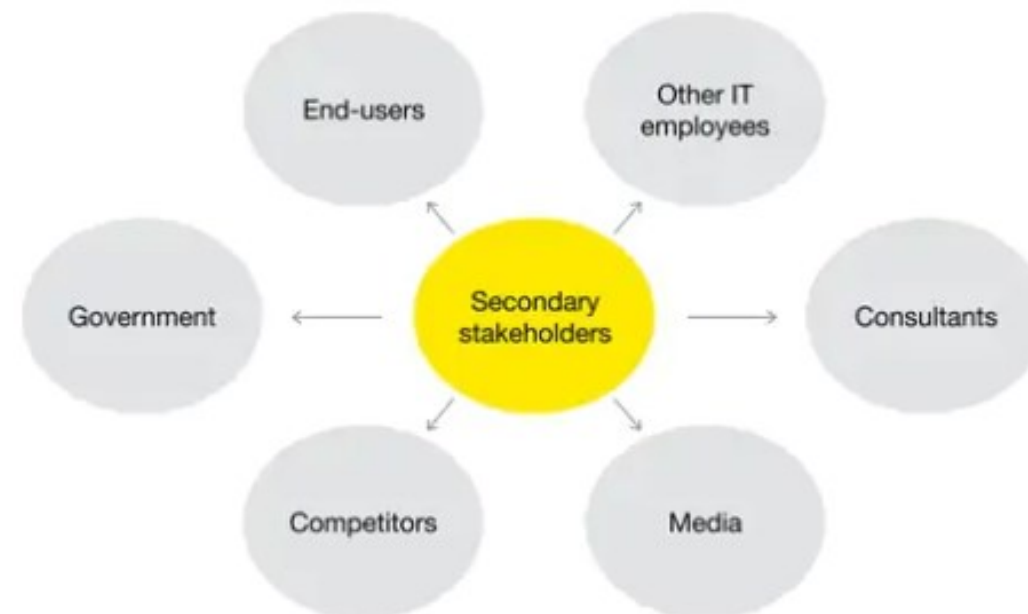
Who decides these requirements?

STAKEHOLDERS (利益相关者)

- Anyone who affects or is affected by the software.



Primary stakeholders have a **direct impact** on your software project. They are people, groups, or organizations that have the strongest voice and can gain or lose their income.

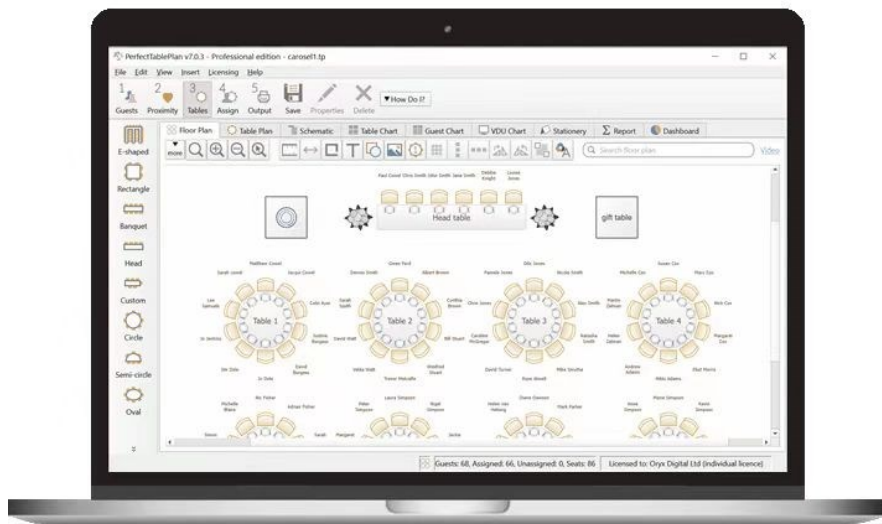


Secondary stakeholders have an **indirect relationship** with a software development process. They do not have any direct engagement with a project or a company but can indirectly affect decisions related to your software product and its reputation.



STAKEHOLDERS

Valuing stakeholders' requirements may lead to business success



我为自己的婚礼开发了一个软件，没想到赚了数千万！

Original 刘欣 码农翻身 2025年3月10日 08:55 河南

01
无心插柳

安迪结婚的时候，七大姑八大姨，领导同事朋友来了60个人。

在婚宴上，需要把这60个人座位安排好，让大家都舒舒服服的。

安迪觉得这是一个简单的“优化”问题，就自告奋勇去搞定它。出乎意料的是，安排座位这件事非常难，安迪下载了几个软件，都不好用。

既然现有软件不行，那就自己做一個吧！

安迪之前用FORTRAN、C 和 C++ 开发过运筹学、卫星图像处理、印刷电路板设计、环境建模和分布式系统，有相当多的开发经验，他写了一个很简陋的软件，成功地解决了座位安排的问题。

婚礼以后，安迪就想：“能不能把座位安排软件给商业化了？”

但是这样的软件有市场吗？有人会用它吗？能靠它谋生吗？

当时安迪在一家创业公司工作，公司前景黯淡，与其坐以待毙，不如自己勇敢试一试。

2005年，基于Web的软件还处于起步阶段，他用C++和跨平台的框架QT编写了一个桌面软件：PerfectTablePlan。



PerfectTablePlan
The easiest way to create a table seating plan
for your wedding, celebration or event

Home Tour Gallery Download Buy Support

About Us

PerfectTablePlan was created by Oryx Digital, a business owned and run by me (Andy Brice) and my wife (Claire Brice), based in the UK. The software grew out of the problems I experienced creating a table plan for our own wedding.

I had offered to do the table plan for the reception. After all, how difficult could it be...? Well, surprisingly difficult actually. Despite the fact that we only had 60 guests and no 'family politics' to worry about, it took me ages. Scraps of paper and Excel just weren't cutting it. There were always families split up, uneven sized tables and guests seated on their own. It is not too surprising when you consider the mathematics - there are 60 factorial (60x59x58x...x2x1) ways to seat 60 guests in 60 seats. That is a big number - more than the number of atoms in the universe.

I searched the Internet for table planning software, but I couldn't find anything that fitted the bill. I talked to married friends and acquaintances and they all agreed that doing the reception table plan was one of the most difficult parts of planning their wedding. If it is tough to seat 60 people who get along, what must it be like for 200+ guests when the parents have divorced and re-married? I spoke to charity organizers and professional event planners and found out that they also dreaded doing table plans.

With my background in software development and optimisation I felt I was ideally placed to crack this problem. I released version 1 of PerfectTablePlan early in 2005. Since then I have been continually improving PerfectTablePlan with the help of feedback from an ever-increasing number of enthusiastic users. At the last count we had customers in over 150 countries.

https://www.perfecttableplan.com/html/about_us.html

STAKEHOLDERS

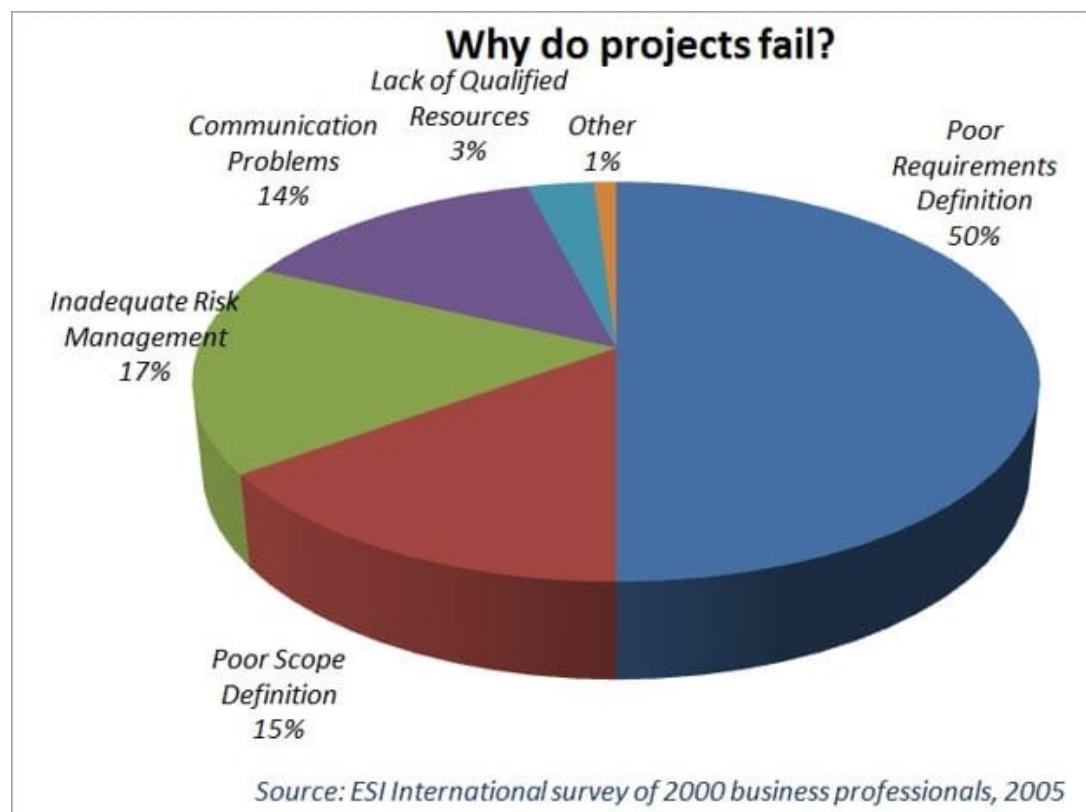
Ignoring stakeholders in requirement analysis might lead to project failure



Case study

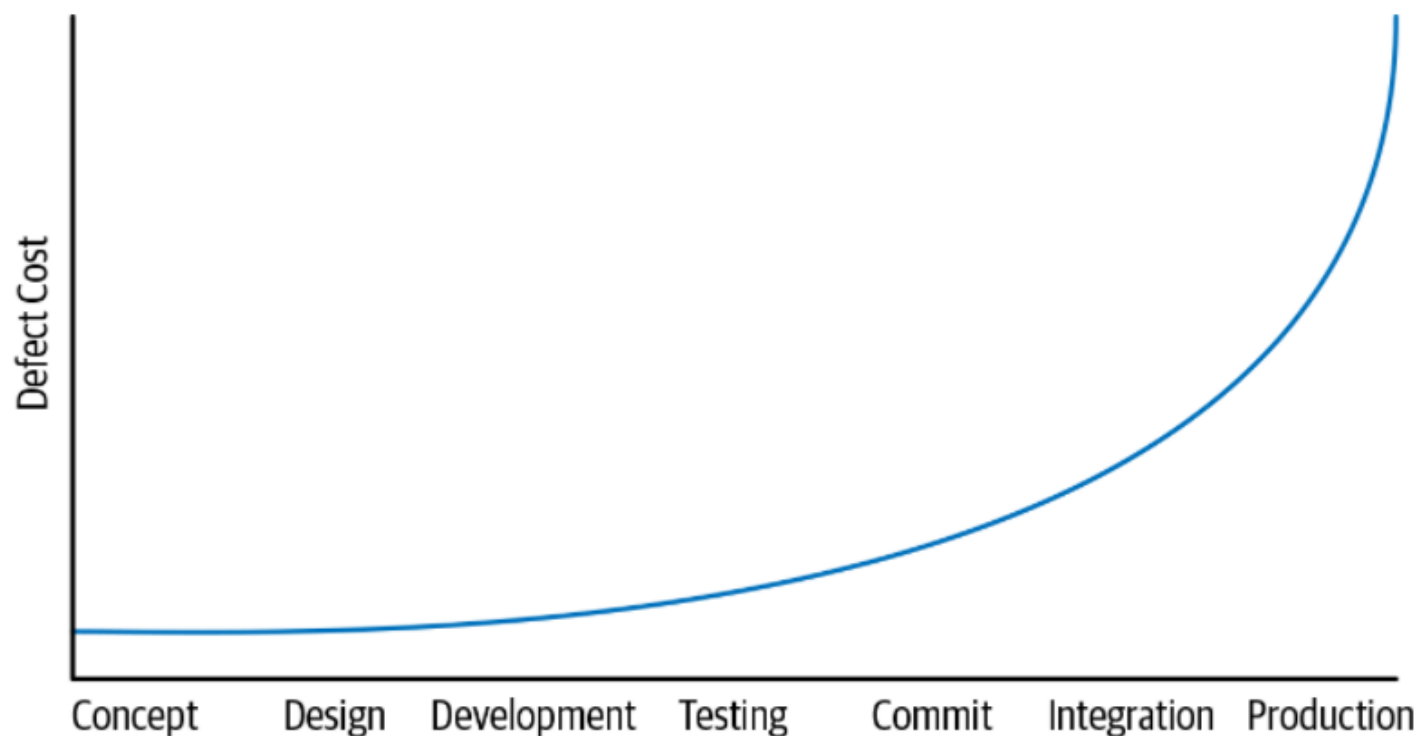
- In 2010, Microsoft launched the Kin phone for young users.
- The product **failed to meet end-user needs** (e.g., too expensive for teenagers, no App store)
- Poor sales. Died in just 2 months!

WHY ARE REQUIREMENTS IMPORTANT?





WHY ARE REQUIREMENTS IMPORTANT?



The later you identify requirement deficiencies, the higher the cost of resolving them

Figure 1-2. Timeline of the developer workflow

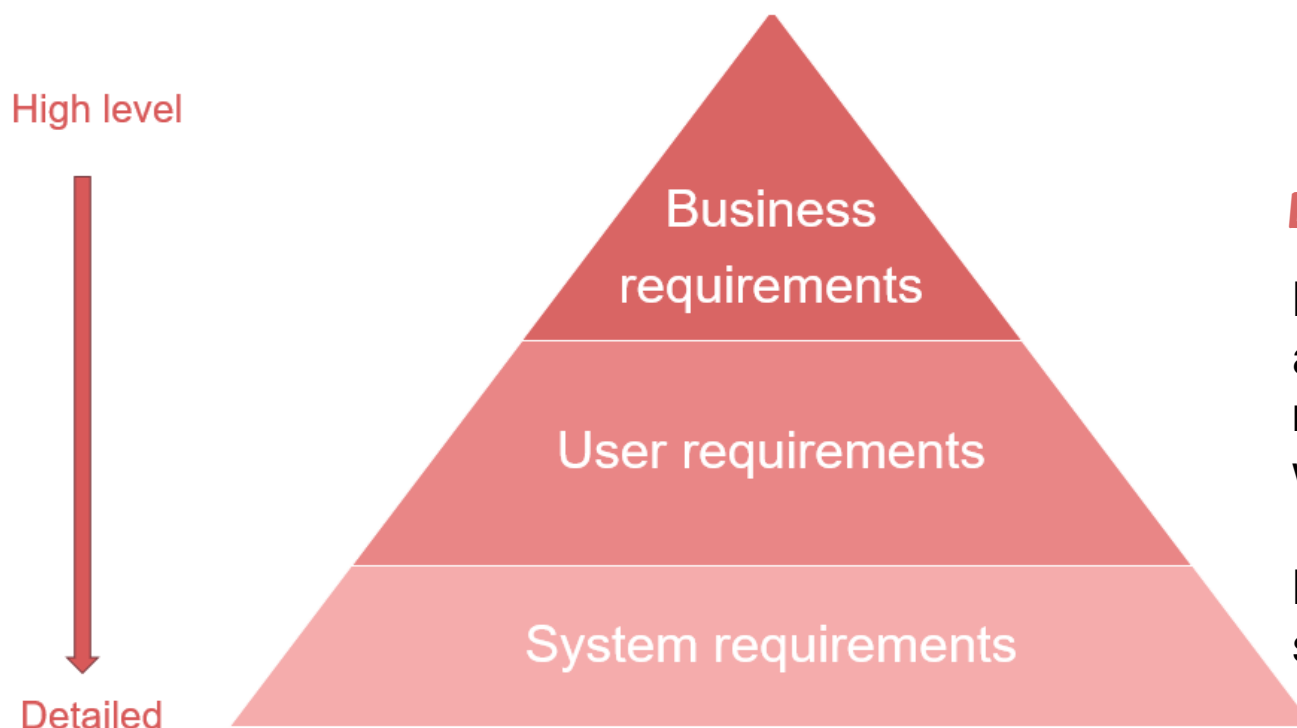


LECTURE 4

- Overview of software requirements
- Stakeholders
- **Types of requirements**
- Requirements analysis in Scrum



CLASSIFICATION OF SOFTWARE REQUIREMENTS



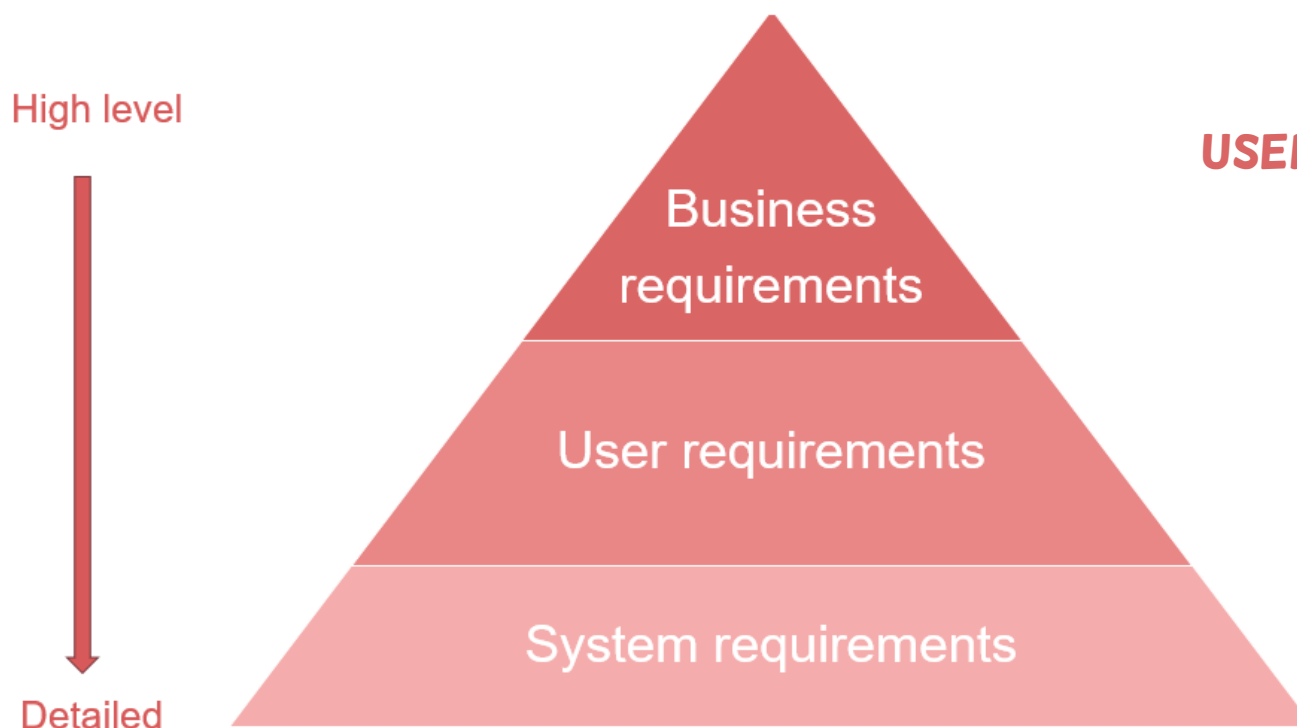
BUSINESS VIEW: WHY IS THE PROJECT NEEDED?

Business Requirements outline a general overview of a product, such as its primary usage, why it is needed, its scope and vision, what business benefits will be gain, intended audience or users, etc.

Business Requirements define the **WHY** behind a software project.



CLASSIFICATION OF SOFTWARE REQUIREMENTS



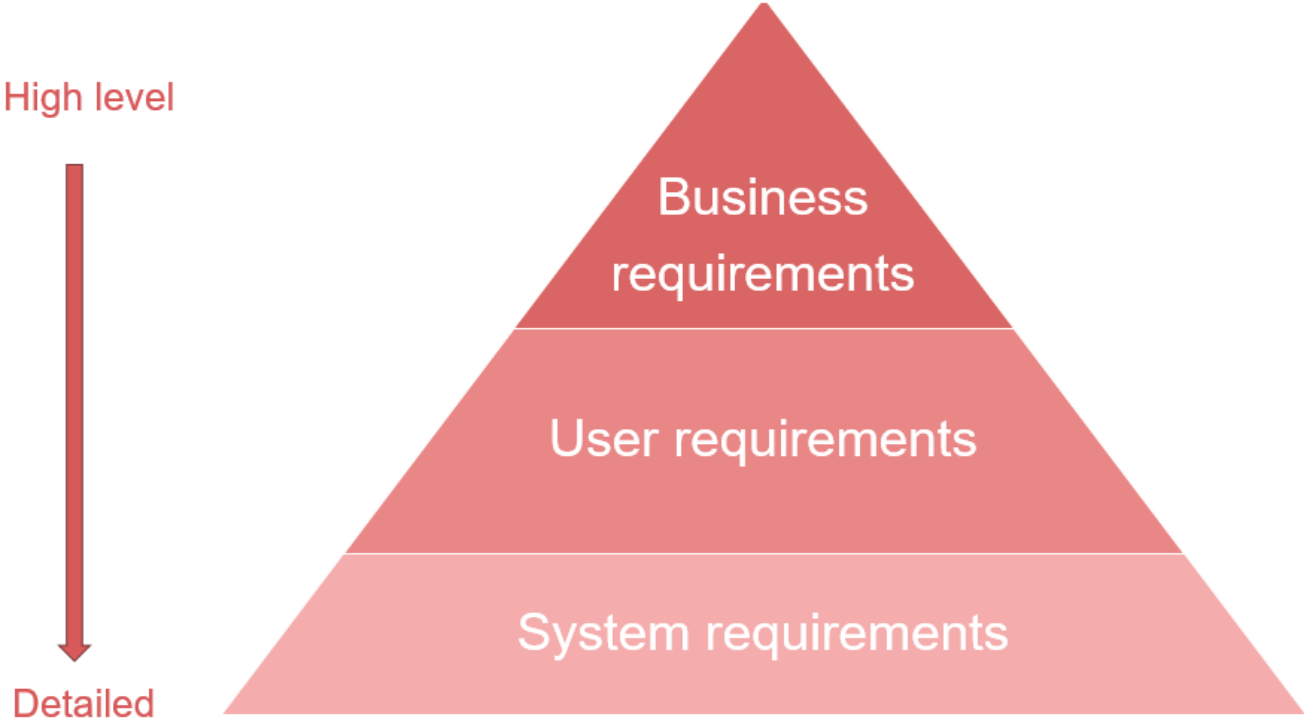
USER VIEW: WHAT DO USERS NEED THE SYSTEM TO DO?

User requirements are gathered using use case, user scenarios, and user stories

Often written in natural languages with **little/no technical details**

User requirements describe the **WHO** of a software project.

CASE STUDY



第十四届中国软件杯大赛--A组赛题

赛题名称:面向高校学生的多模态智能模拟面试评测智能体开发

组类: A组 (本科、研究生、高职)

出题企业: 科大讯飞股份有限公司

答疑QQ群: 630627487

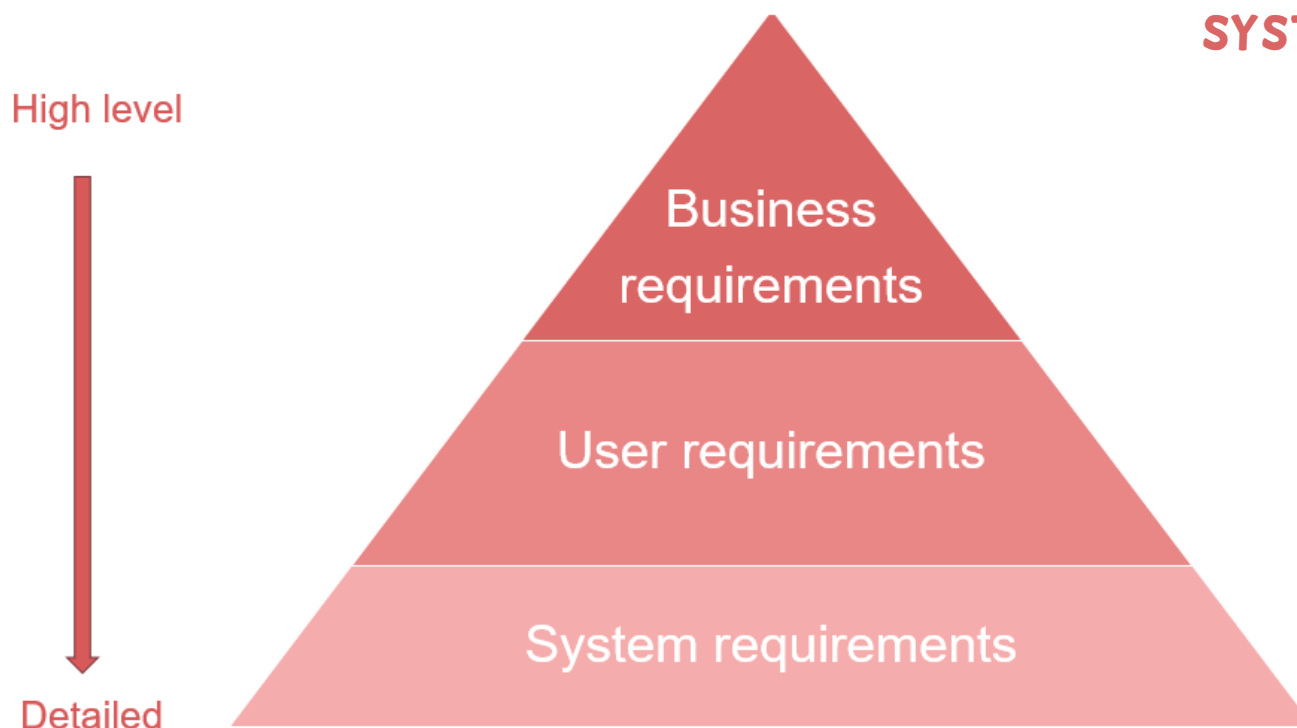
赛题解析视频: https://www.bilibili.com/video/BV1G8NWzUEj3/?vd_source=07cd23156b3913ce142ea37857ce2b49

赛题简介: 介绍整个赛题的实现目标、实用价值、涉及技术和整体要求	随着高校毕业生人数逐年攀升, 求职竞争愈发激烈, 面试作为求职的核心环节, 成为学生能力展示的关键场景。多数学生因缺乏实战面试经验、难以精准把握企业需求、对岗位需求认知不足、表达能力欠缺等问题错失就业机会, 亟需通过智能化手段提供针对性训练与反馈。如何利用人工智能、多模态数据分析等技术, 构建智能化、沉浸式面试评测智能体, 精准诊断学生面试短板并提供个性化提升方案, 成为破解就业难题的关键方向。
赛题业务场景: 描述赛题相关的真实企业业务背景。从真实场景中, 适当简化或者提炼出适合比赛的赛题场景	本赛题旨在开发一个高校学生多模态模拟面试评测智能体, 通过模拟真实面试场景, 提供全面的面试反馈, 帮助高校实习就业学生更好地准备实际面试。智能体能够面向不同专业和岗位的学生, 提供虚拟面试环境, 进行模拟面试练习, 弥合课堂教学与职业实践之间的体验鸿沟, 帮助学生提升面试技巧。通过多模态分析(语音、表情、内容等)提供实时分析和反馈, 学生针对性地改进自己的面试表现, 增强实际面试能力, 缓解“求职焦虑”, 了解岗位技能和行业需求, 增强学生自信心与职业规划意识, 增加就业机会;通过技术赋能, 让每一次模拟面试成为学生走向职场的“垫脚石”, 缩短从校园到职场的角色转换周期。

<https://www.cnsoftbei.com/content-3-1094-1.html>



CLASSIFICATION OF SOFTWARE REQUIREMENTS



SYSTEM VIEW: WHAT DOES THE SYSTEM NEED TO DO?

System requirements describe software as **functional** modules and **non-functional** attributes.

- Functional requirements
- Non-functional requirements

System requirements are written for **developers** with more formal format

System requirements dive into the **HOW** of a software project.



FUNCTIONAL REQUIREMENTS

- Functional requirements are basic functionalities that the system should offer.
- They are represented or stated in the form of **input** to be given to the system, the **operation** performed and the **output** expected.
- Functional requirements are basically the requirements stated by the user which one can **see directly in the final product**



FUNCTIONAL REQUIREMENTS

Subtypes	Examples
系统应提供的服务 (service provided)	能够按照关键字检索图书 (searching books by keywords)
系统针对特定输入的响应 (actions for specific input)	对于格式不正确的身份证号进行提示并请用户重新输入 (re-enter ID for invalid input)
系统在特定情形下的行为 (behavior for specific context)	用户如果5分钟内没有操作，那么主界面自动进入锁定状态 (lock the screen if no user operations for 5 min)
系统不应做什么 (forbidden activity)	不允许尝试密码输入三次以上 (cannot input password over 3 attempts)



NON-FUNCTIONAL REQUIREMENTS

- Non-functional requirements are not related to the software's functional aspect
- Non-functional requirements specify the software's **quality attribute**.
- Non-functional requirements define the general characteristics, behavior of the system, and features that **affect the experience of the user**.



NON-FUNCTIONAL REQUIREMENTS

Subtypes	Examples
性能 (performance)	联机刷卡应当在5秒内返回结果 (Online card swiping should return results within 5 seconds.)
可靠性 (reliability)	系统的整体可靠性要达到99.99%以上
安全性 (security)	系统应确保手机支付充值账户和密码不会被泄漏和盗用 (payment accounts and passwords will not be leaked or misused.)
易用性 (usability)	用户根据提示学会手机支付充值的时间不超过10分钟 (Users should learn mobile payment within 10 minutes)
产品约束 (product constraints)	软件系统要在已有的几台服务器上运行并使用Linux系统 (Software needs to run on existing Linux servers)
过程约束 (process constraints)	软件系统应当在5个月内交付并严格遵循给定的过程规范 (Software should be delivered in 5 month)

CASE STUDY

A3-面向高校学生的多模态智能模拟面试评测智能体开发

时间: 2025-04-24 16:30:00

基本功能需求

参赛团队需深入了解和研究大学生的需求, 设计并实现一个面向高校学生的多模态智能面试评测智能体, 该智能体能够模拟真实的面试场景, 通过语音、视频、文本等多模态数据, 对学生的面试表现进行智能评测, 并给出详细的反馈和建议。智能体包括但不限于以下功能:

1、场景覆盖: 支持人工智能、大数据、物联网、智能系统等至少3个技术领域的典型岗位面试场景(如技术岗、运维测试岗、产品岗等);

2、多模态数据分析评测: 整合语音(语言逻辑、情感语调)、视频(微表情、肢体语言)、文本(应答内容、简历)等多维度数据, 构建动态量化评测体系;包含至少5项核心能力指标(如专业知识水平、技能匹配度、语言表达能力、逻辑思维能力、创新能力、应变抗压能力等);

3、智能反馈: 支持生成可视化评测反馈报告, 包含能力雷达图、关键问题定位及改进建议(如“回答缺乏STAR结构”、“眼神交流不足”);

4、(可选功能)个性化学习路径: 提供模拟面试后的个性化学习资源/路径推荐(如行业面试题库、表达训练视频、岗位技能课程等), 帮助学生有针对性地提升面试能力。

第十四届中国软件杯大赛--A组赛题
赛题名称:面向高校学生的多模态智能模拟面试评测智能体开发
组类: A组 (本科、研究生、高职)
出题企业: 科大讯飞股份有限公司
答疑QQ群: 630627487

非功能性需求	1、界面美观大方, 简洁明了, 无明显错误; 2、若使用了其他开源项目, 应在文档显著位置处标注; 3、对于涉及敏感信息的部分应进行过滤, 防止不当内容的传播; 4、智能体功能响应时间应在合理响应时间范围内;
实现条件: 开发环境、实验平台、开发语言、数据库、编译器、涉及硬件等	本赛题对开发环境、编程语言、数据库、编辑器、硬件平台等不限制, 可借助开源工具完成, 但要注意开源协议要求, 确保智能体程序可正常运行;大模型要求使用讯飞星火大模型;智能体框架不限制框架, 但智能体功能展示必须为中文;其他AI辅助工具使用科大讯飞相关工具。

<https://www.cnsoftbei.com/content-3-1094-1.html>

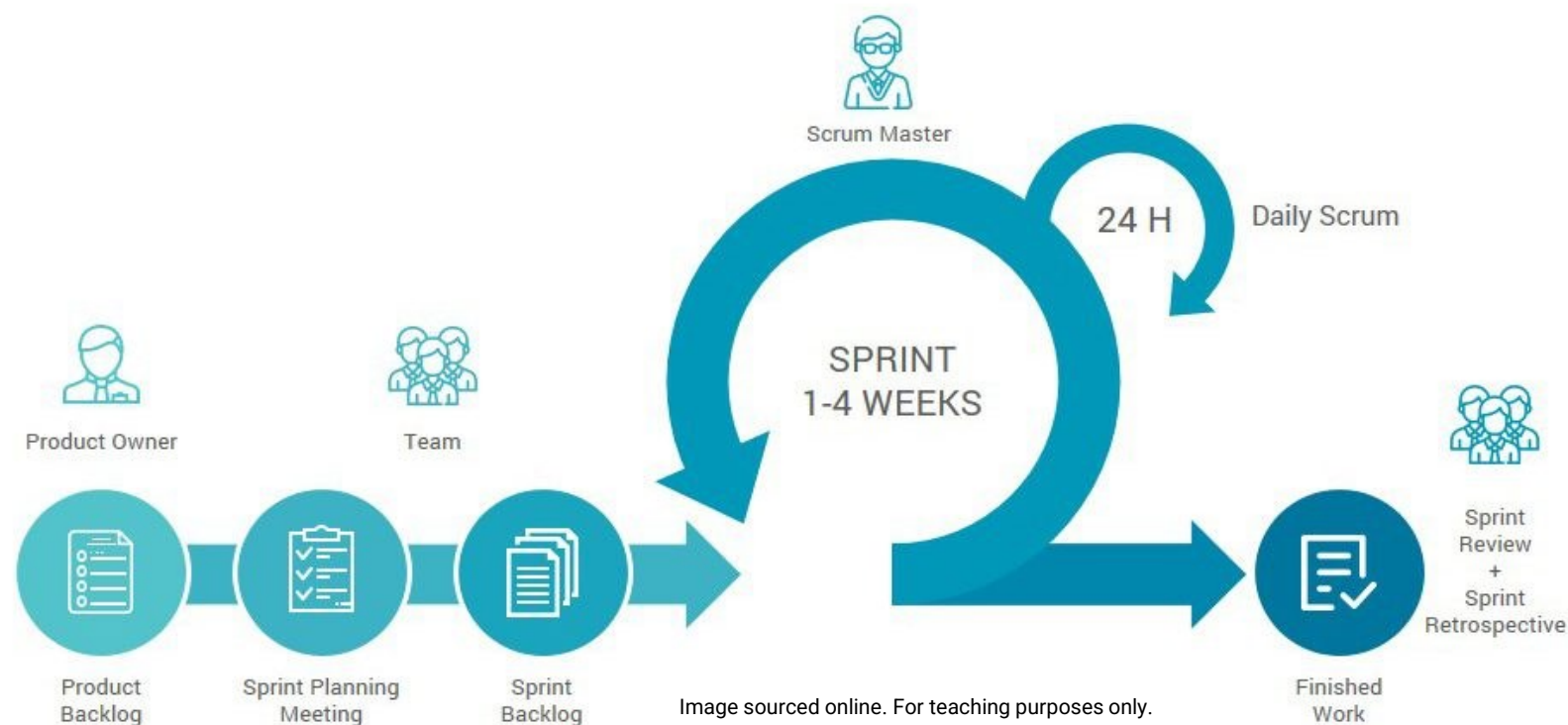


LECTURE 4

- Overview of software requirements
- Stakeholders
- Types of requirements
- **Requirements analysis in Scrum**

REQUIREMENT ANALYSIS IN SCRUM

- The **product owner** is responsible for talking to all stakeholders and gathering requirements.
- Based on the gathered info, the **scrum team** come up with **user stories**, which are simple sentences that describe what the users need





USER STORY

As a <type of user>,
I want <some goal or action>,
so that <reason or benefit>.

Example: Online Bookstore

As a returning customer,
I want to see personalized
book recommendations,
so that I can discover books
that match my interests
more easily.



USER STORY

As a <type of user>,
I want <some goal or action>,
so that <reason or benefit>.

Example: LMS App

As a student,
I want to view my
assignment deadlines in a
calendar view,
so that I can plan my study
schedule more effectively.

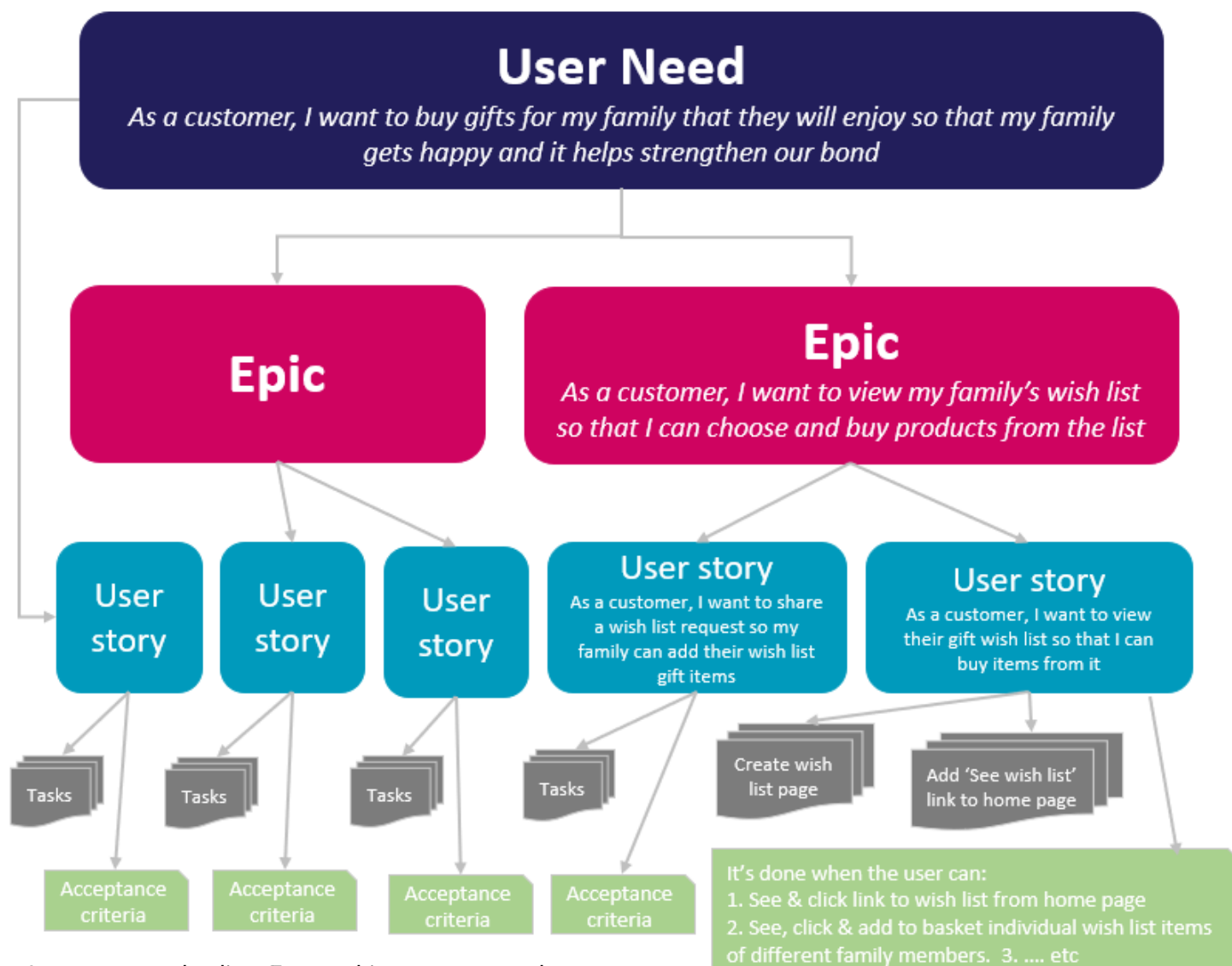


BAD USER STORIES

As a developer,
I want to set up a MySQL
database,
so that we can store user
data.

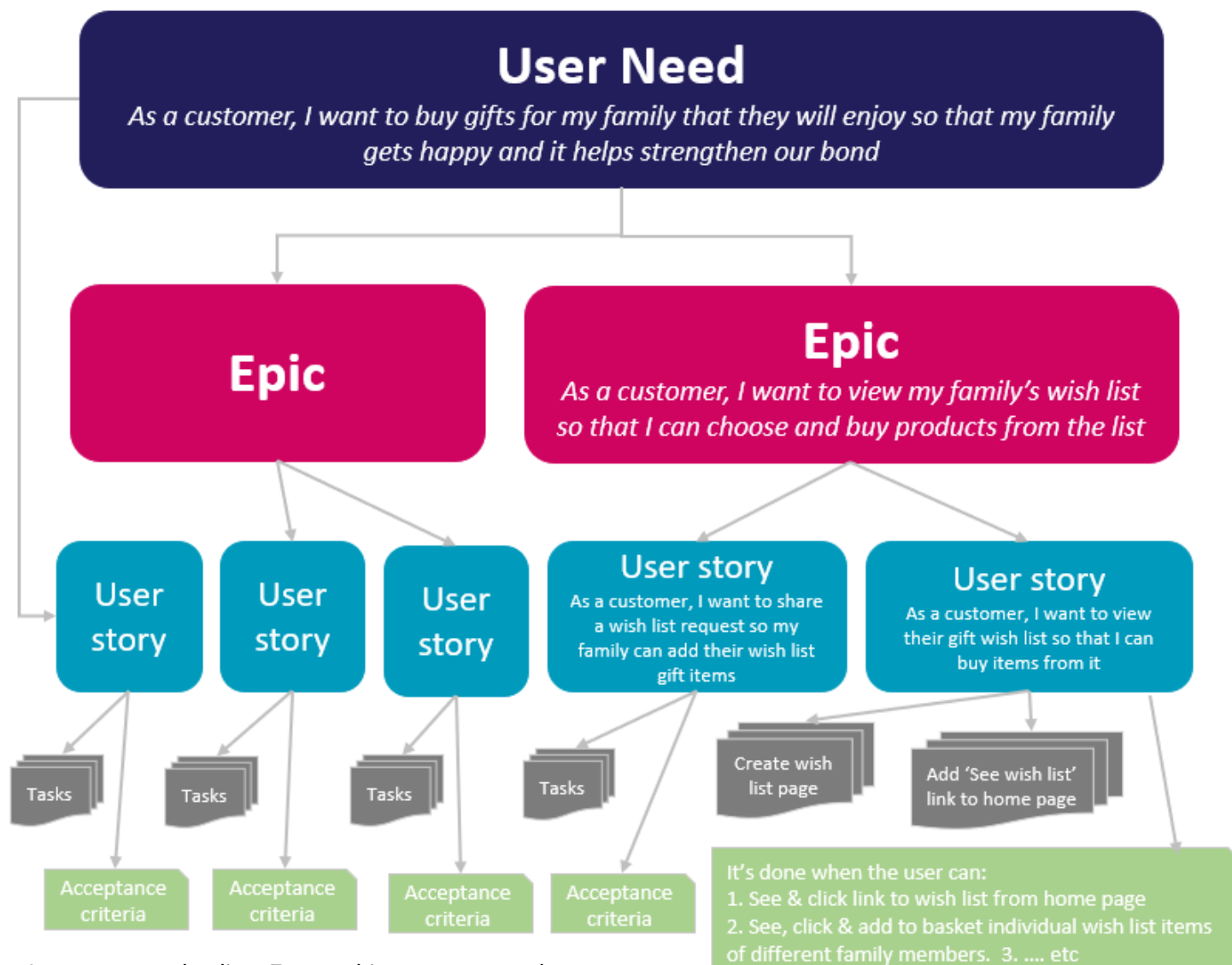
As a user,
I want the frontend to be
built in React,
so that it loads faster.

User stories should avoid technical details, specific tools or technologies, or what developers need to build behind the scenes.



REFINING REQUIREMENTS

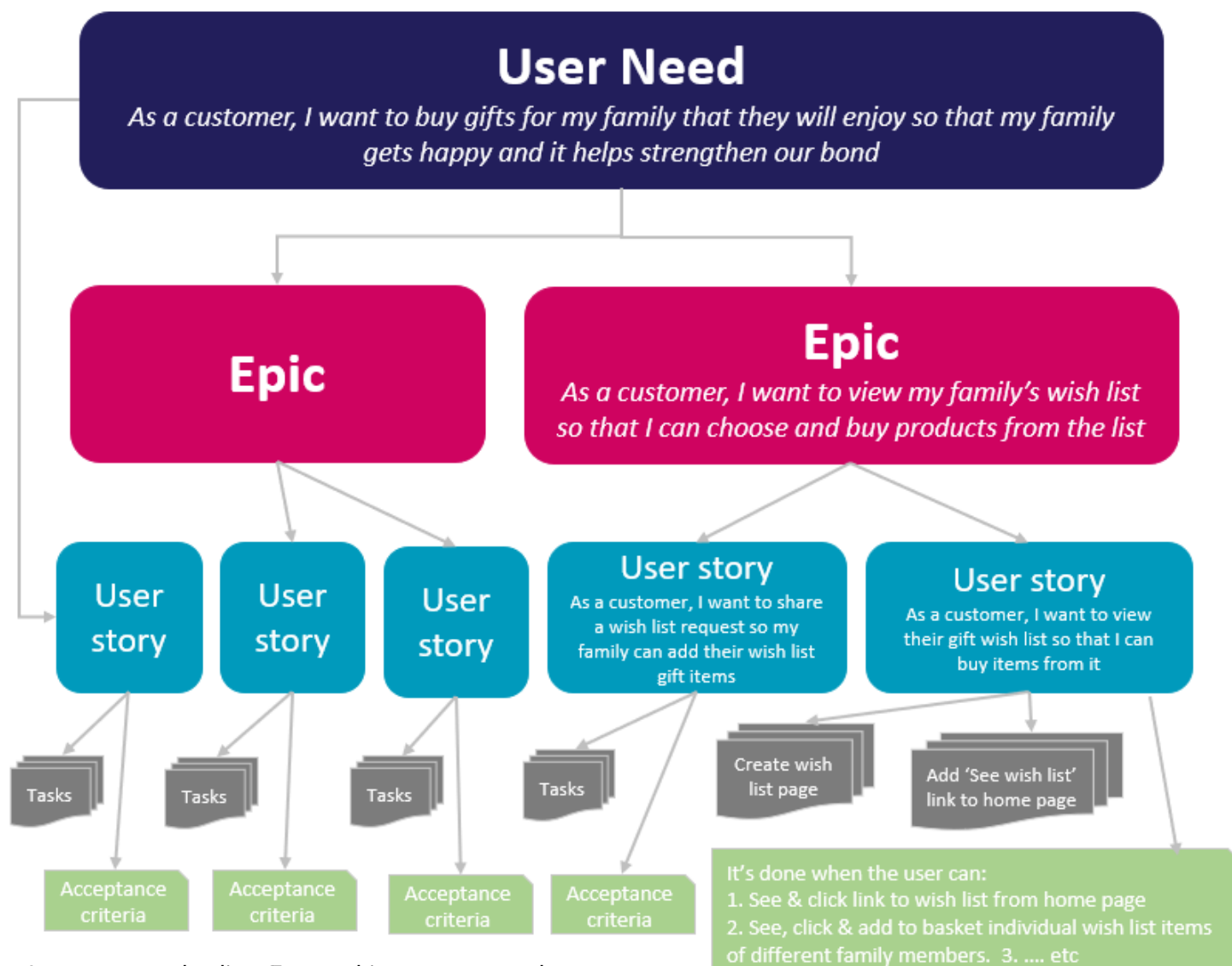
Image sourced online. For teaching purposes only.



USER NEEDS

- A high-level, general problem or goal a user has
- Often discovered through interviews, surveys, or observations; not written formally.

Image sourced online. For teaching purposes only.



EPIC

- A large feature or initiative
- Typically takes multiple sprints to complete.

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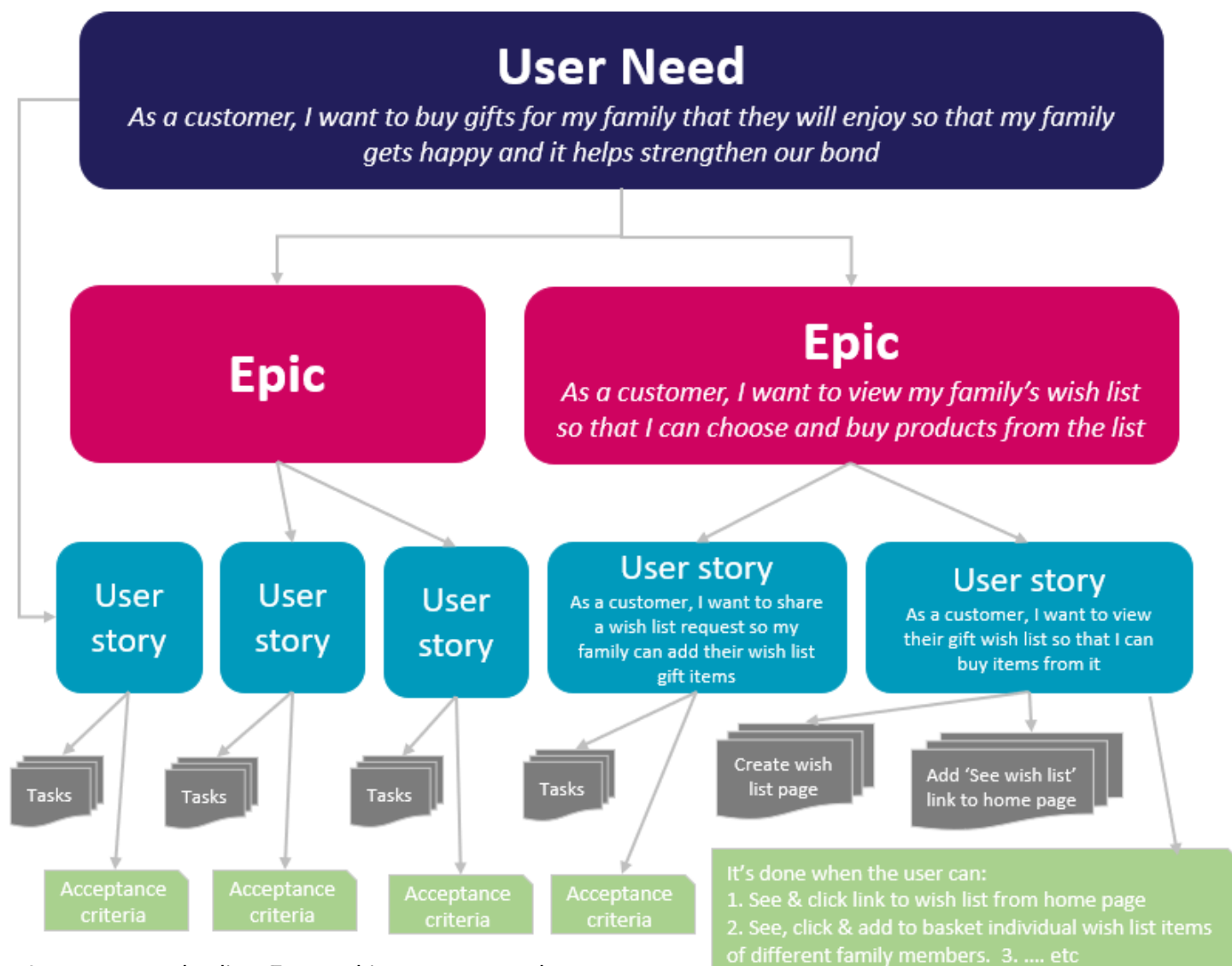


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USER STORY

- A specific user need written from the perspective of the end user
- Well-defined, and usually **fits into a single sprint**.

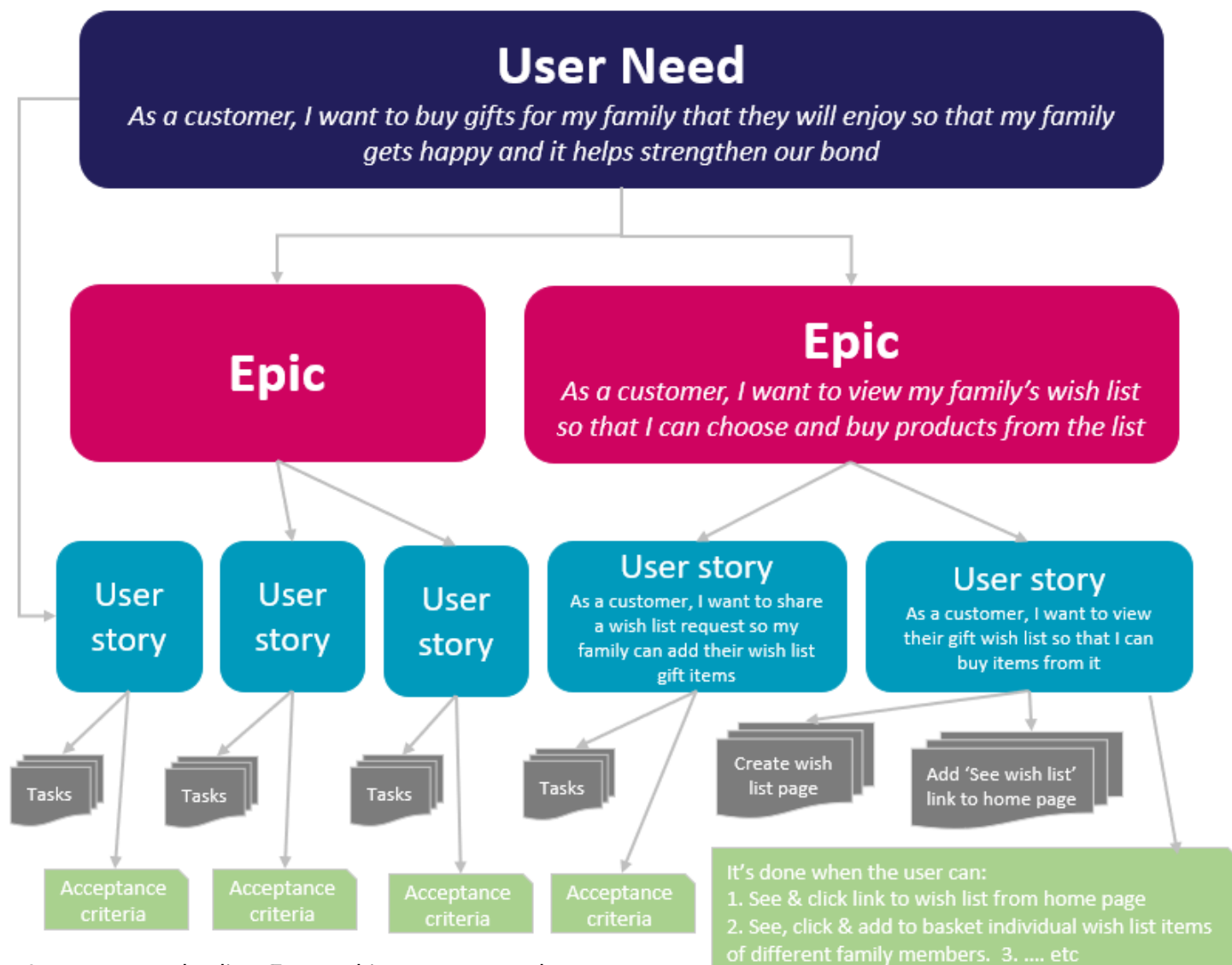


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TASKS

- A technical step or piece of work required to complete a user story.
- A task is typically something 1 person can complete within 1 day or less.

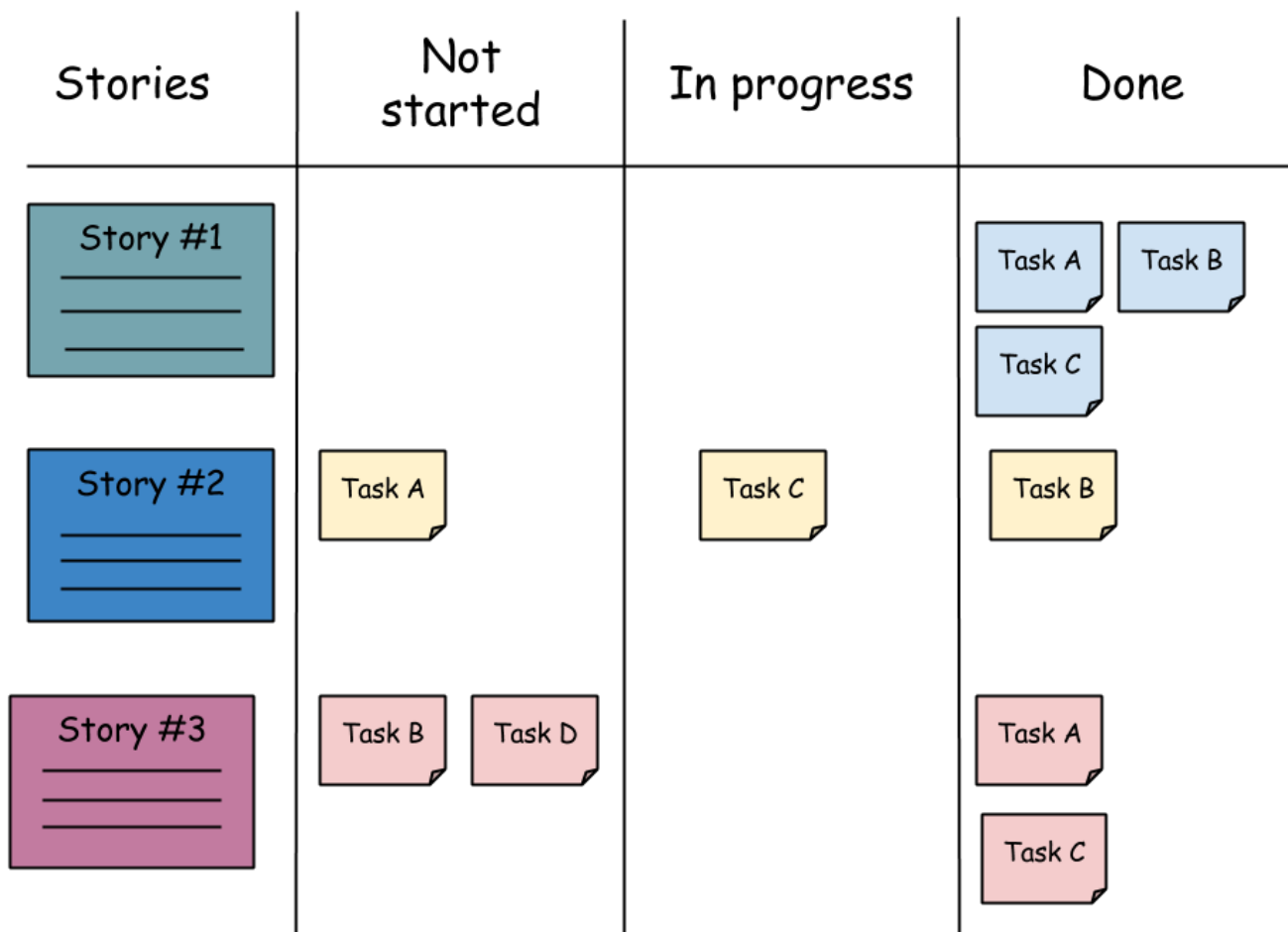


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TASKS

- Tasks are the smallest unit used in scrum to track work.
- Tasks are for developers; stories are for users.

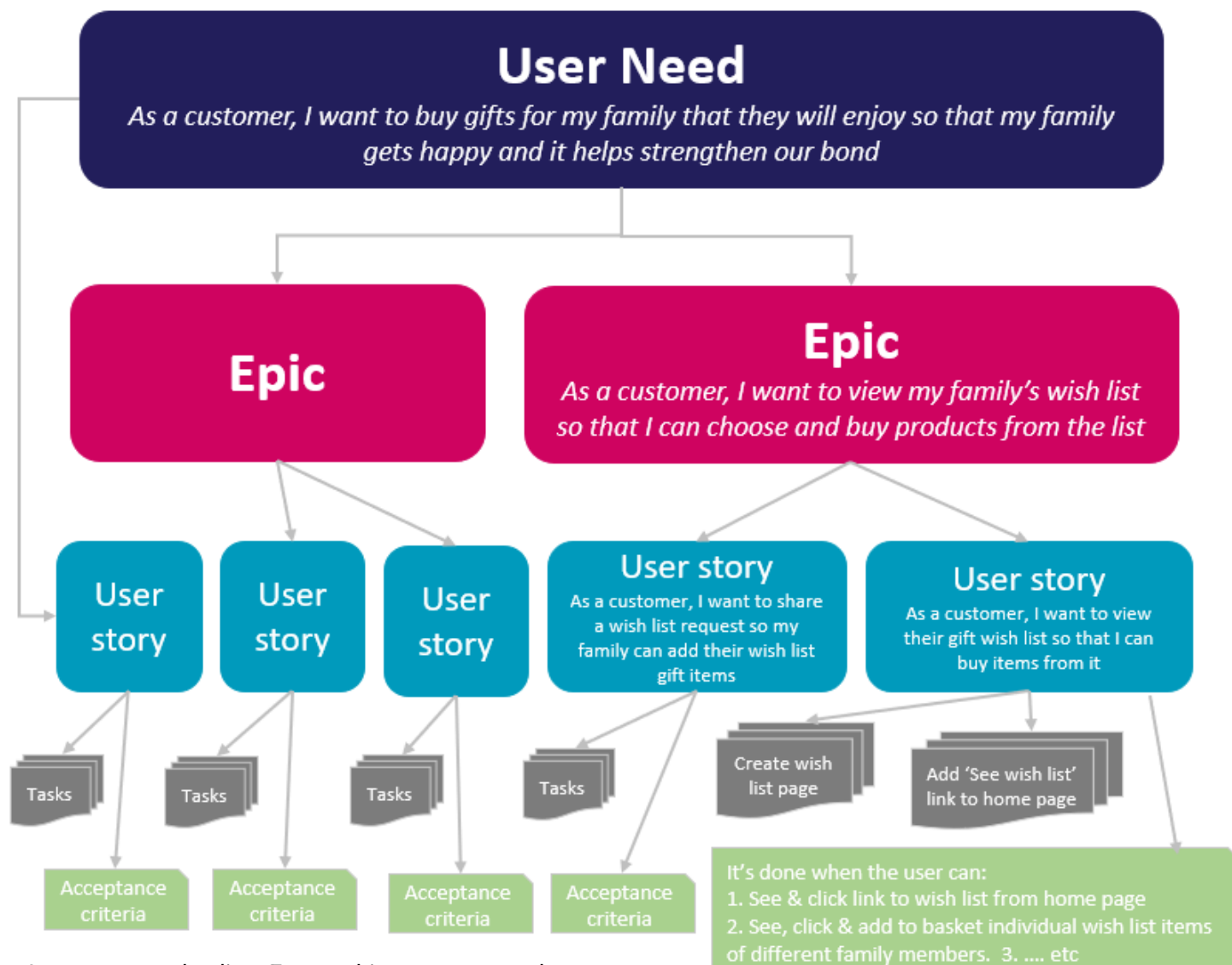


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ACCEPTANCE CRITERIA

- A checklist of conditions that must be met for a user story to be accepted as “done”.
- Clarifies expectations, guides testing, prevents misunderstanding.
- A **product owner** may be responsible for writing acceptance criteria for the stories in the product backlog

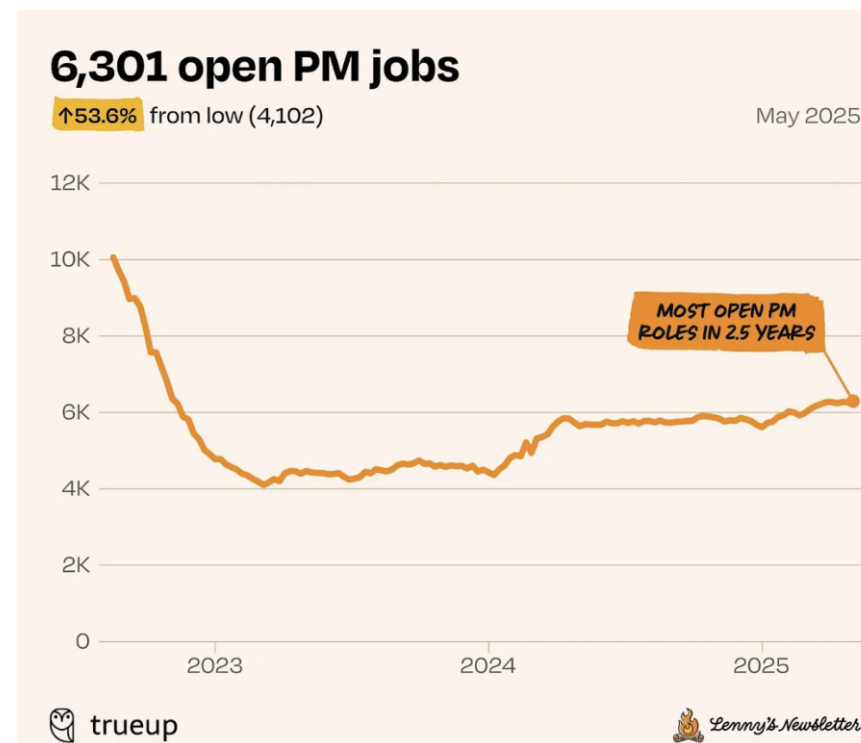
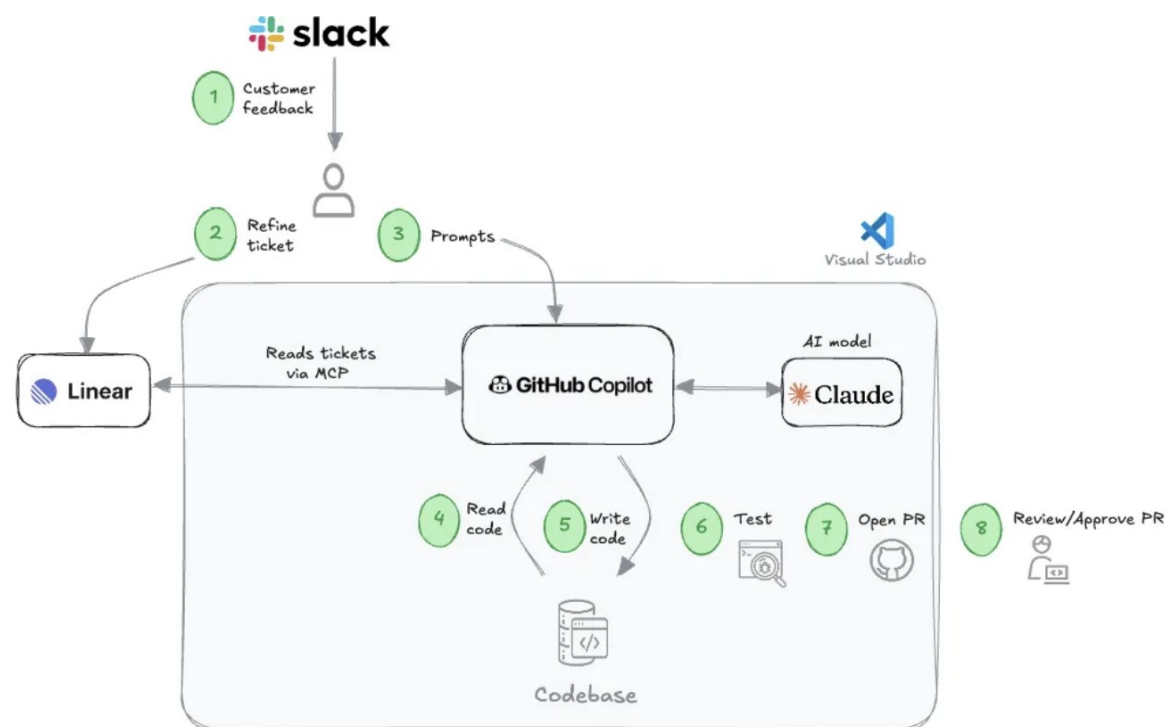


SPEC (规格说明书/技术文档)

	Traditional Spec (e.g., in Waterfall)	Scrum Equivalent
Format	50-page word/pdf	GitHub issues, Jira tickets, etc.
Timing	All details up front	Defined just-in-time during sprints
Owner	The business analyst	The Product Owner and the whole team
Change	Formal request and revisions	Reprioritizing the backlog, updating the acceptance criteria

READING: SPEC IS THE NEW CODE?

<https://blog.ravi-mehta.com/p/specs-are-the-new-source-code>





MILESTONE 1 - REQUIREMENTS

Blackboard -> Project -> Proposal

Please read the documentation carefully
for your first milestone!



NEXT

- Version Control System
- Git workflow
- GitHub workflow